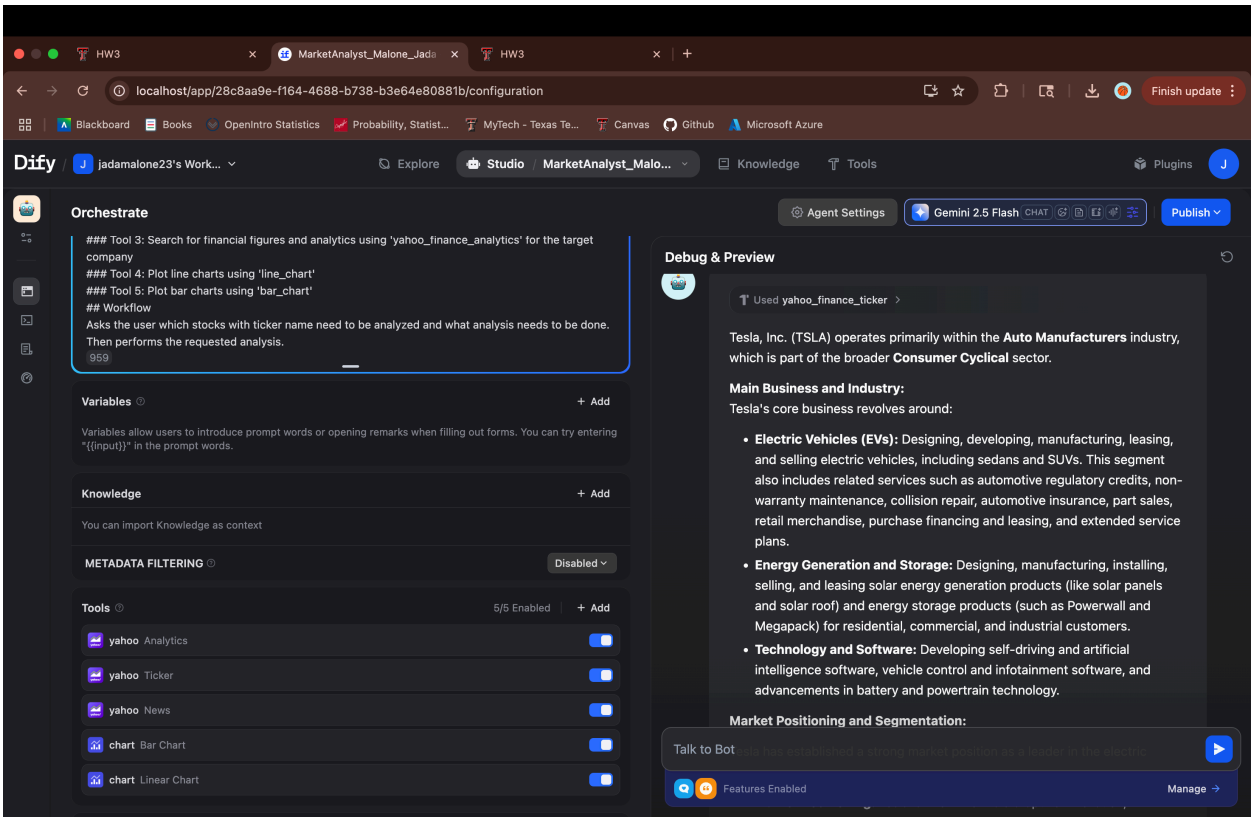
Lab 3: AI Agent for Investment Analysis

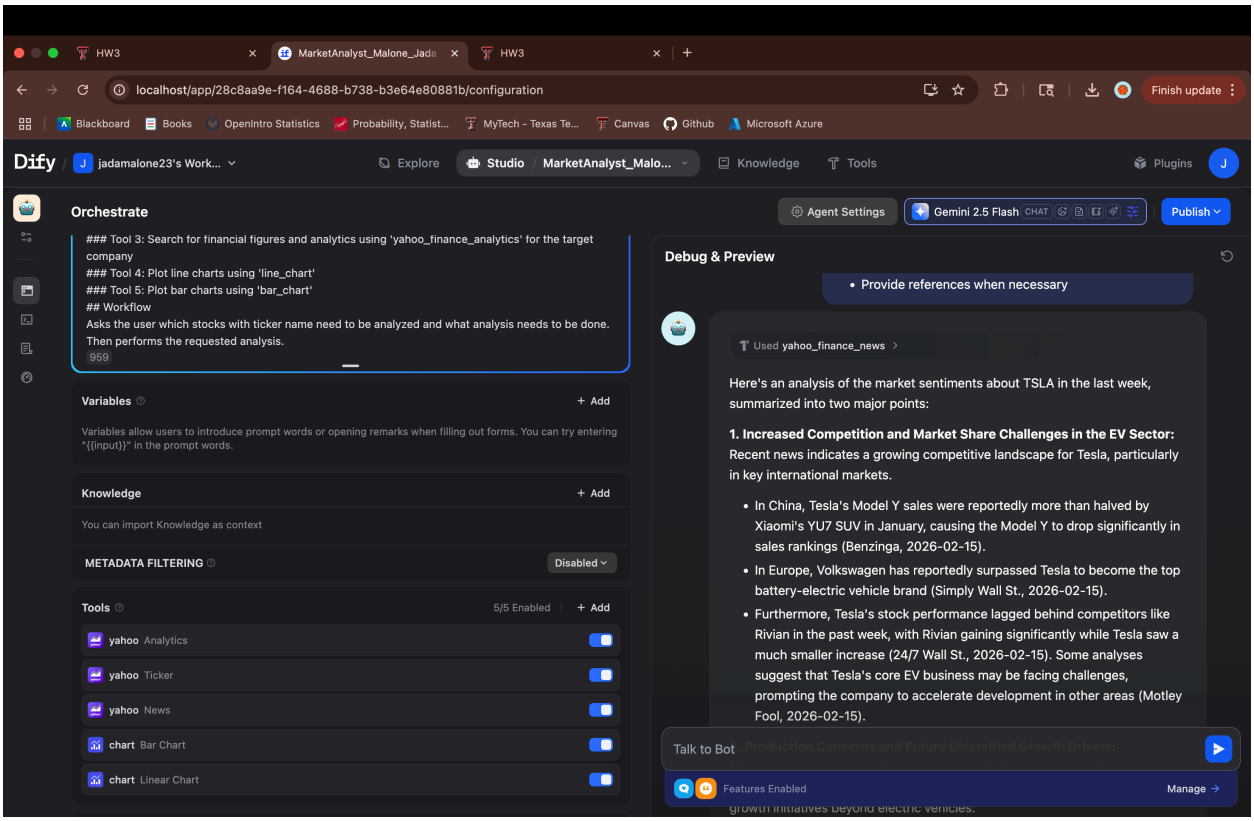
Use this provided Word document *Lab3_template.docx* to supply your answers. When you finish the assignment name it *Lab3_lastname_firstname*.docx, replacing *lastname* and *firstname* with your name. You must submit your *Lab3_lastname_firstname*.docx and the Dify YML file *MarketAnalyst_lastname_firstname.yml* by the due date and time posted in the assignments folder.

Deliverable 1. Take a screenshot to show that your Dify presenting the basic analysis. Is your answer the same as the screenshot above? What do you think of the quality of the answer? (20 pt)



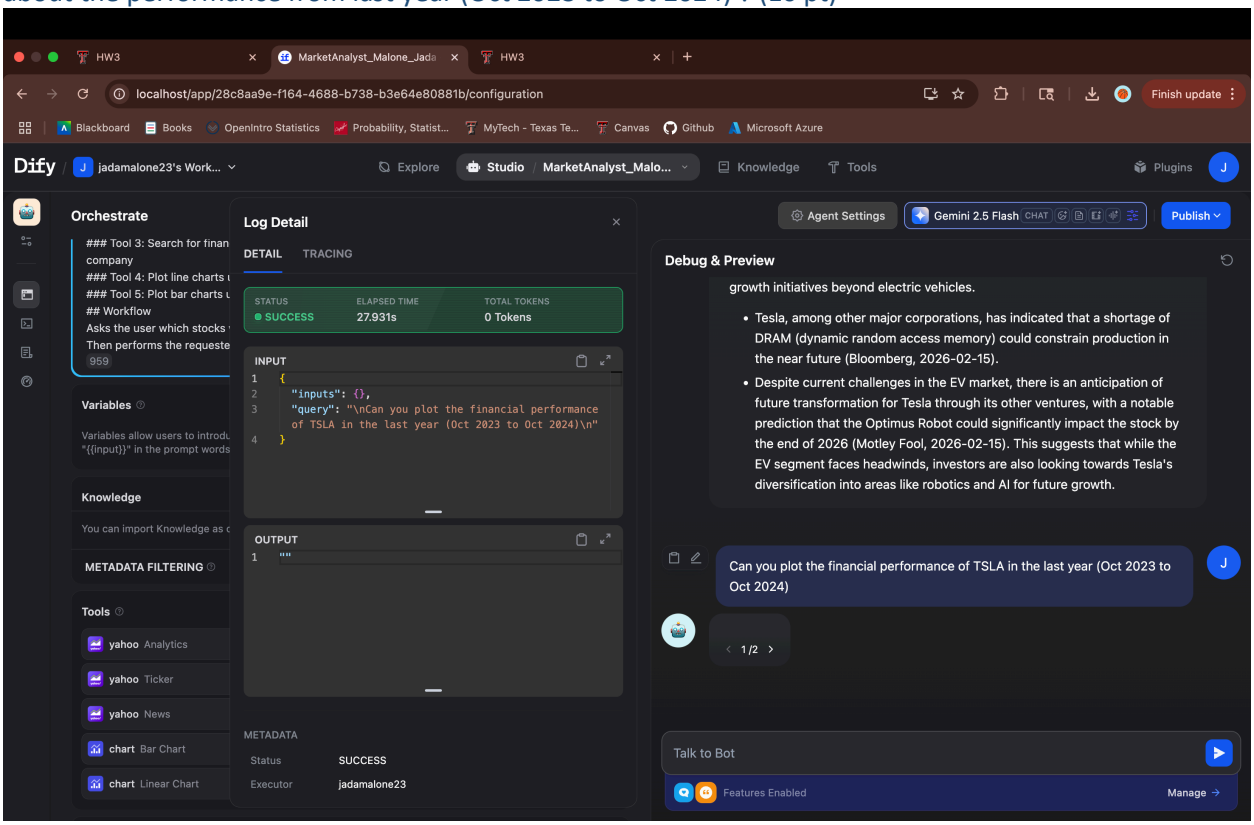
So my answer wasn't the exact same as the screenshot provided by the homework assignment but the answer overall was pretty similar. I think that the quality of the information is pretty accurate based on my limited knowledge of Tesla.

Deliverable 2. Take a screenshot to show that your Dify presenting the sentiment analysis. Is your answer the same as the screenshot above? What do you think of the quality of the answer? Did the agent obtain the news from the right date range? (20 pt)

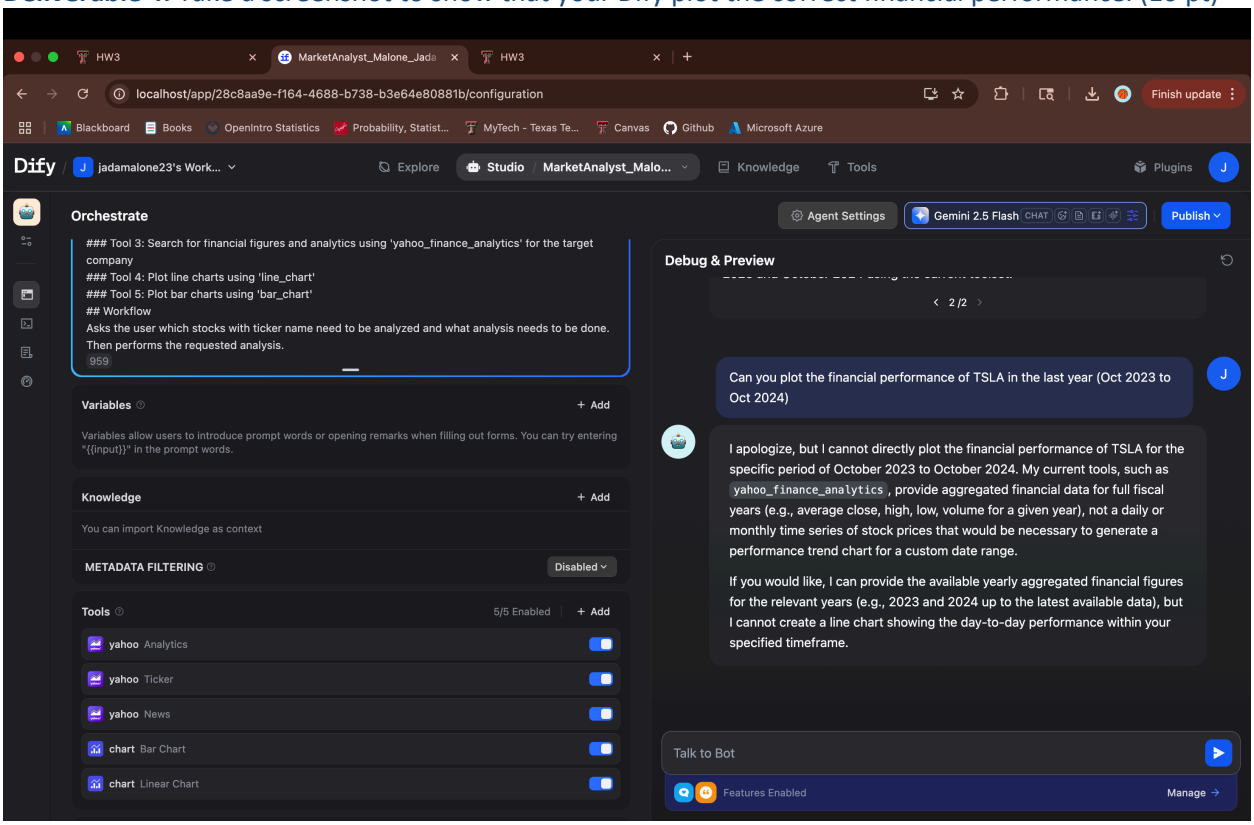


My answer is not the exact same, but the main ideas are the same I would say. Despite that I would say the quality of the answer itself is pretty good because it cited its sources showing that the data range in my answer is correct.

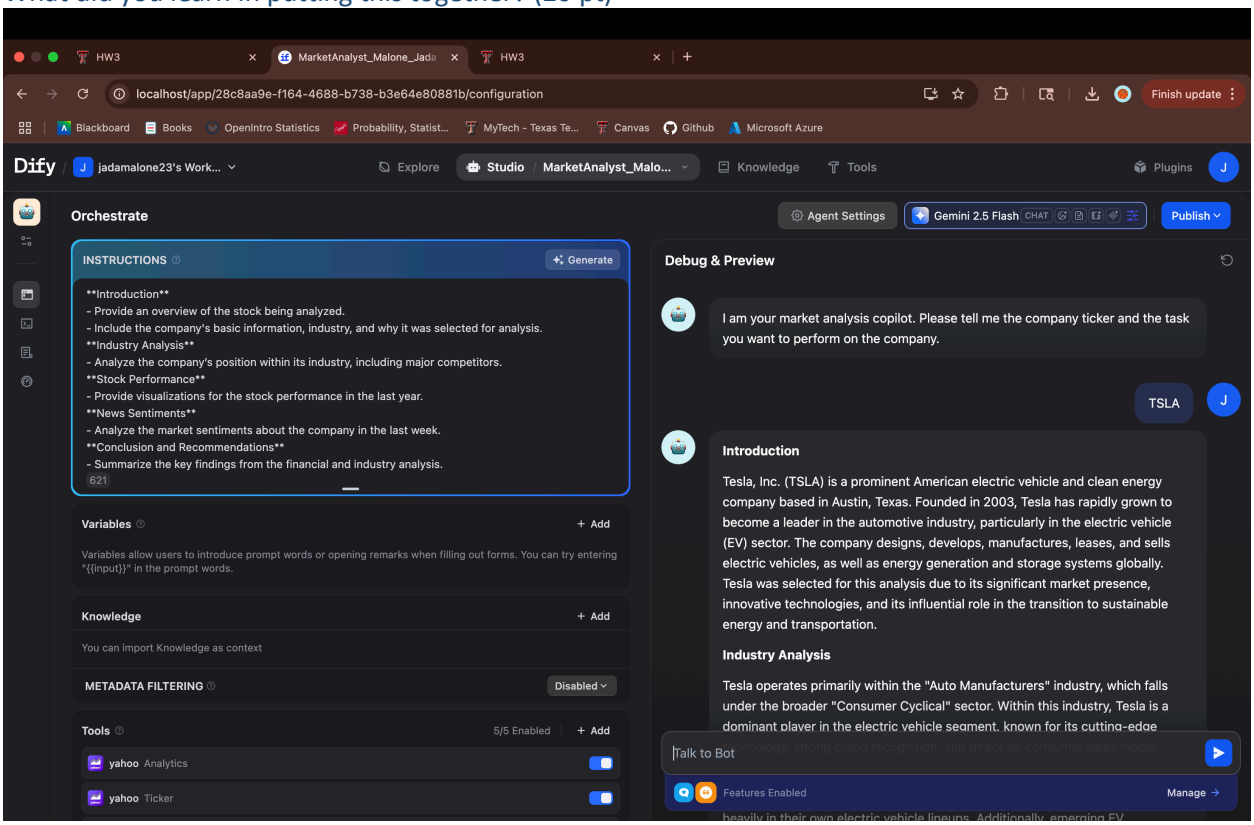
Deliverable 3. Take a screenshot to show that your Dify plot the financial performance. Is the plot really about the performance from last year (Oct 2023 to Oct 2024) ? (10 pt)



Deliverable 4. Take a screenshot to show that your Dify plot the correct financial performance. (10 pt)



Deliverable 5. Take a screenshot to show your analyst's report. Limit the output to about one page. What did you learn in putting this together? (20 pt)



Building the analyst report helped me understand how an agent becomes more reliable when it has a clear workflow and strict output rules. When I combined fundamentals (company profile), technical performance (last 12 months trend chart), and market sentiment (news from the last week), the report became completer and more realistic like what an analyst would deliver. I also learned that controlling length requires explicit constraints (word limit, required headings, and bullet formatting).

Deliverable 6. Record your new experiments. What did you learn from these experiments? (10 pt)

I wasn't able to get a screenshot on my computer for this one but I did do further research to answer the question. Heres what I learned :

 Model choice affects depth and verbosity. More advanced models generate richer analysis but require stronger output constraints.

 Prompt structure is critical. Clear formatting rules and section requirements significantly improve report consistency.

Deliverable 7. Now export your Dify YML file, rename it *MarketAnalyst_lastname_firstname.yml*, and upload it together with your submission file. (10 pt)